STA108 Term Project

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Appendix

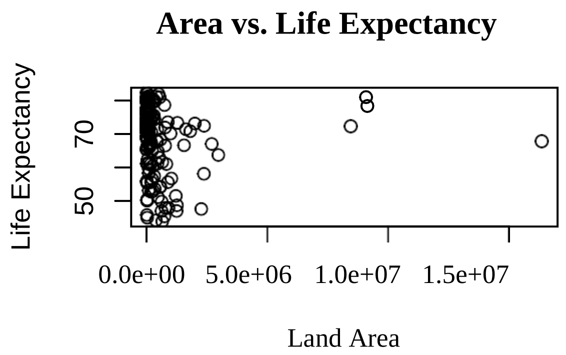
Term Project

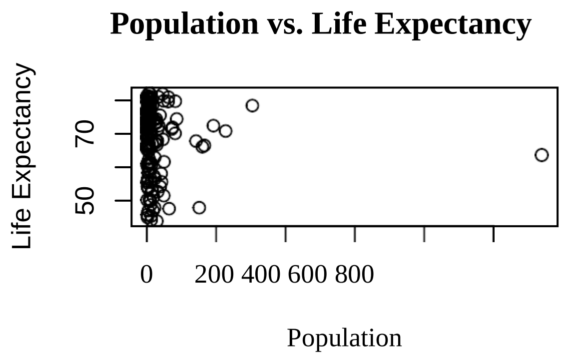
```
#Subset 80% of data
countries_data <- read.csv("/Users/hpimp_girlpower/Downloads/countries.csv")
n_count= nrow(countries_data)
set.seed(10)
subset_id_count= sample(n_count,0.8 *n_count)
count_subset= countries_data[subset_id_count, ]

#Set relevant variables
area=count_subset$LandArea
pop=count_subset$Population
rural=count_subset$Rural
health=count_subset$Health
internet=count_subset$Internet
elderly=count_subset$ElderlyPop
birth=count_subset$BirthRate
life_expect=count_subset$LifeExpectancy
carbon=count_subset$CO2
GDP=count_subset$GDP
cell=count_subset$Cell
```

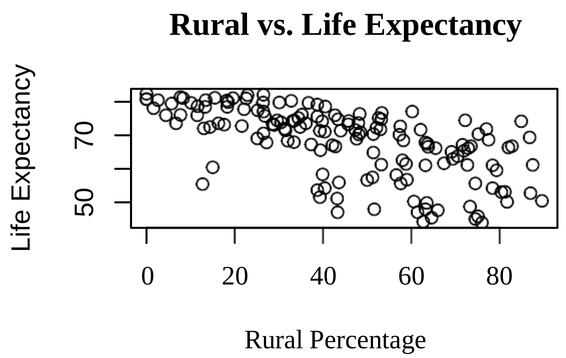
```
#Load needed libraries
library(MASS)
library(leaps)
```

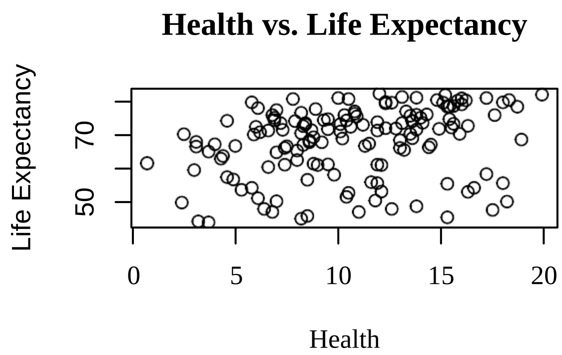
Initial Relationship

```
#Scatterplots for each predictor to determine relationship to outcome variable
par(mfrow=c(2,2))
plot(area, life_expect,xlab= 'Land Area',ylab= 'Life Expectancy', main='Area vs. Life
        Expectancy')
plot(pop,life_expect,xlab= 'Population',ylab= 'Life Expectancy', main='Population vs. Life
        Expectancy')
plot(rural,life_expect,xlab= 'Rural Percentage',ylab= 'Life Expectancy', main='Rural vs. Life
        Expectancy')
plot(health,life_expect,xlab= 'Health',ylab= 'Life Expectancy', main='Health vs. Life
        Expectancy')
```

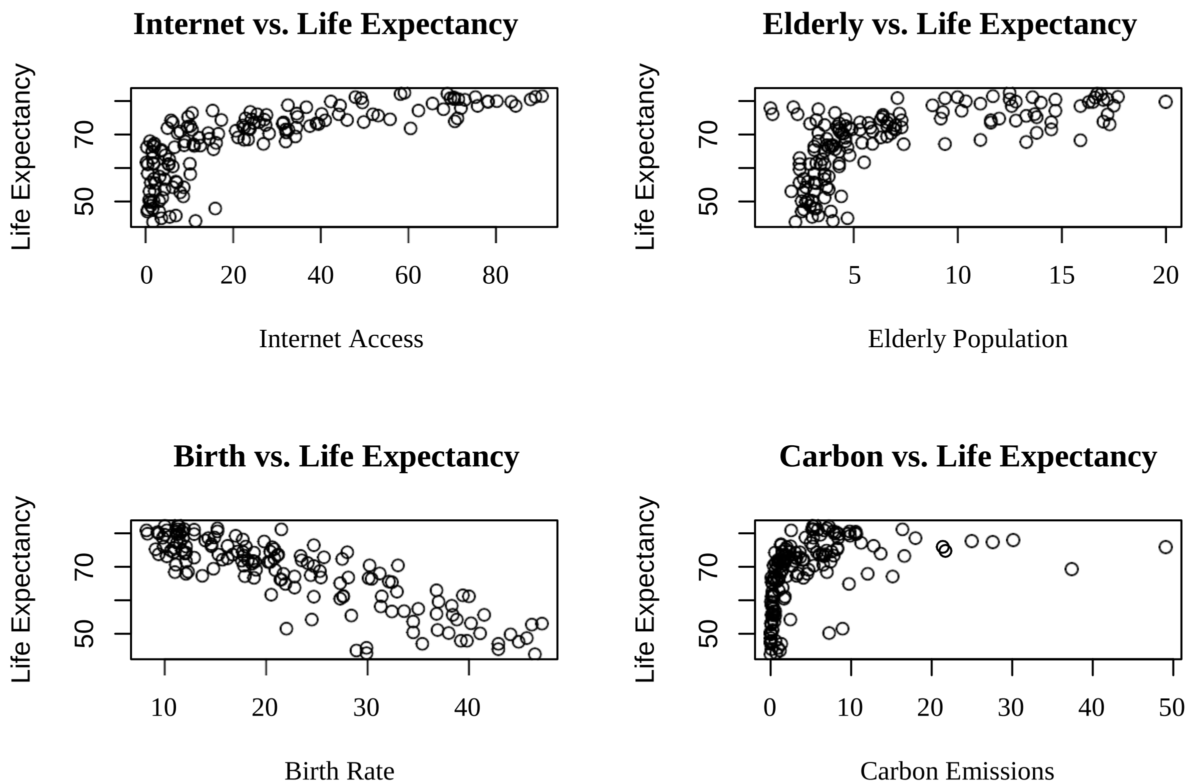




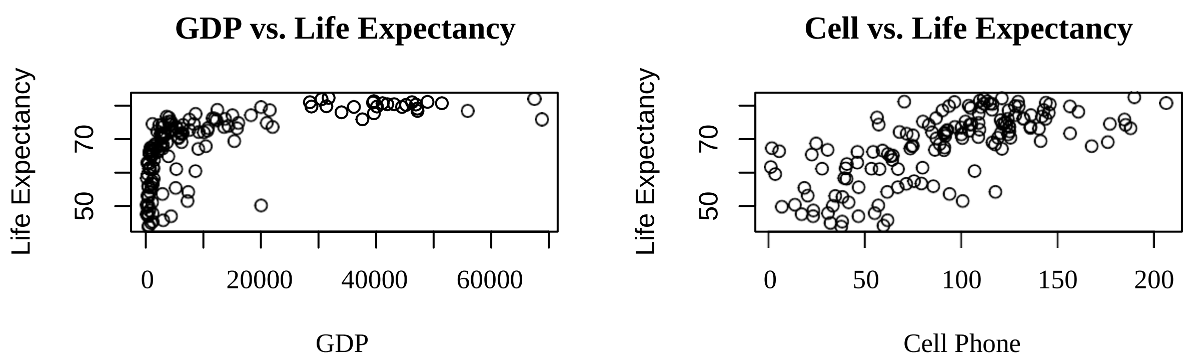





```r
plot(internet,life_expect,xlab= 'Internet Access',ylab= 'Life Expectancy', main='Internet vs. Life
        Expectancy')
plot(elderly,life_expect,xlab= 'Elderly Population',ylab= 'Life Expectancy', main='Elderly vs. Life Expectancy')
plot(birth,life_expect,xlab= 'Birth Rate',ylab= 'Life Expectancy', main='Birth vs. Life
        Expectancy')
plot(carbon,life_expect,xlab= 'Carbon Emissions',ylab= 'Life Expectancy', main='Carbon vs. Life
        Expectancy')
```

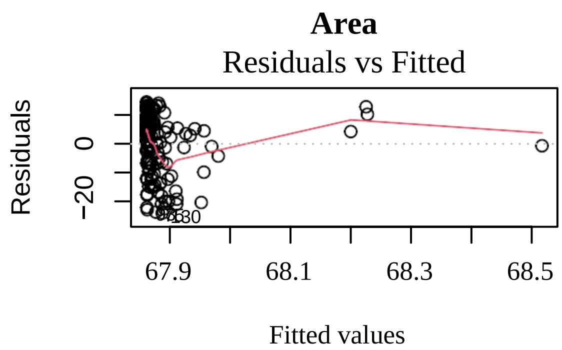


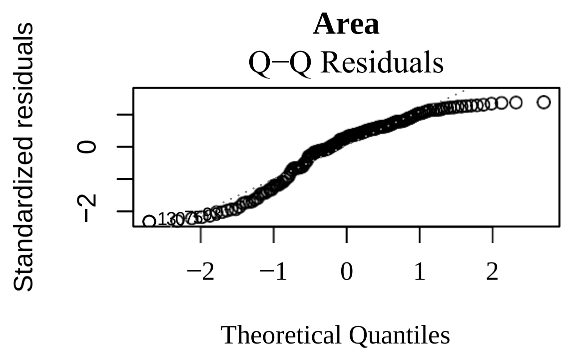

```r
plot(GDP,life_expect,xlab= 'GDP',ylab= 'Life Expectancy', main='GDP vs. Life
     Expectancy')
plot(cell,life_expect,xlab= 'Cell Phone',ylab= 'Life Expectancy', main='Cell vs. Life Expectancy')
```

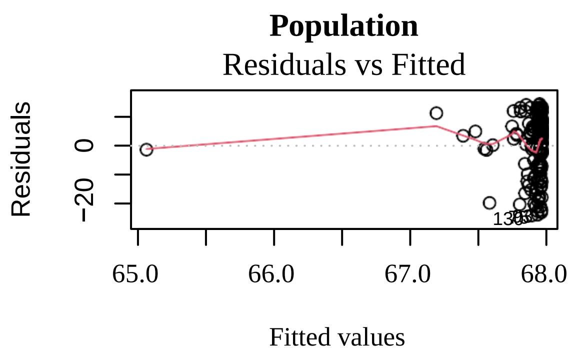


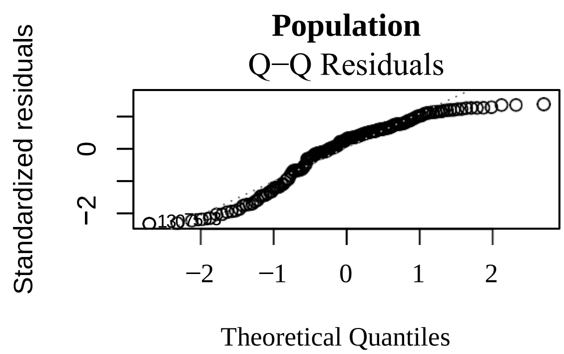
Simple Regression Diagnostics

```
#Run simple linear regressions for each predictor to check for assumption violations

par(mfrow=c(2,2))

#AREA
area_model= lm(life_expect~area)
plot(area_model,which=1,main='Area')
plot(area_model,which=2,main='Area')

#POPULATION
pop_model= lm(life_expect~pop)
plot(pop_model,which=1,main='Population')
plot(pop_model,which=2,main='Population')
```

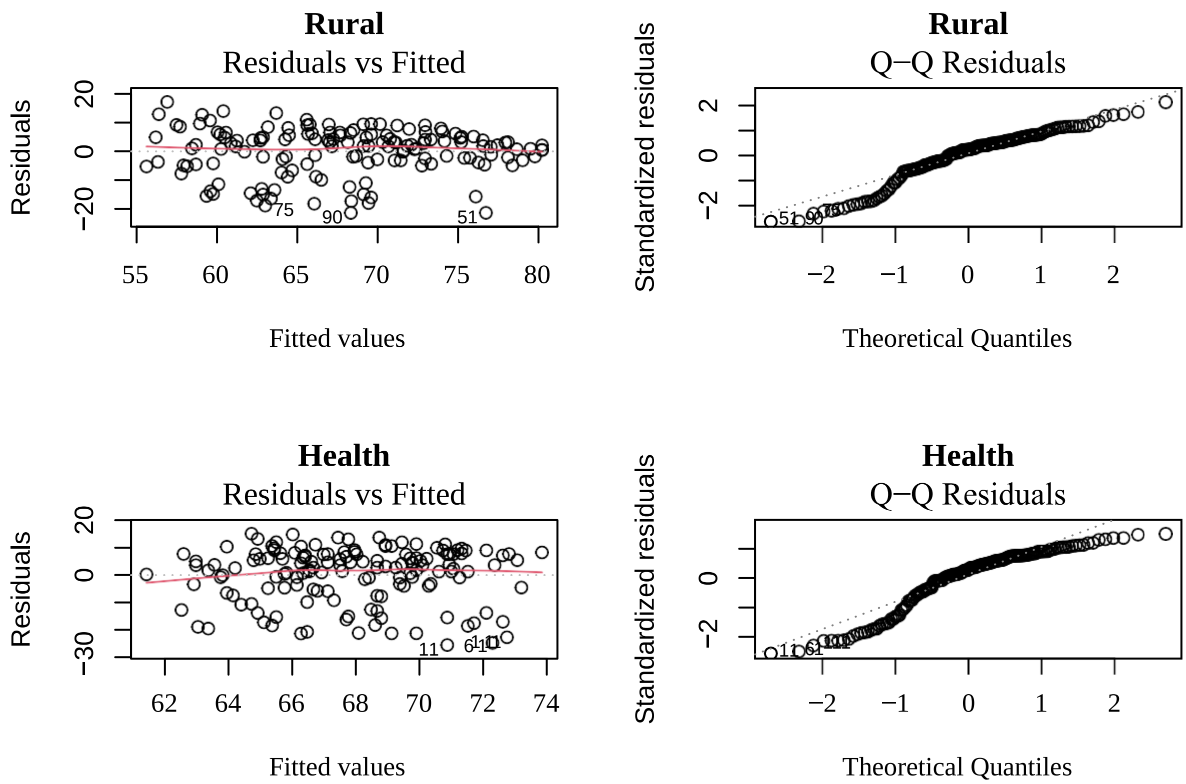




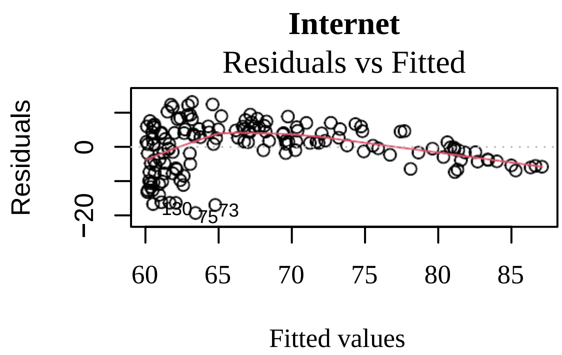


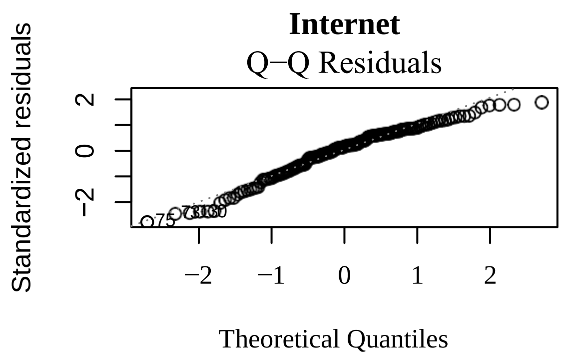


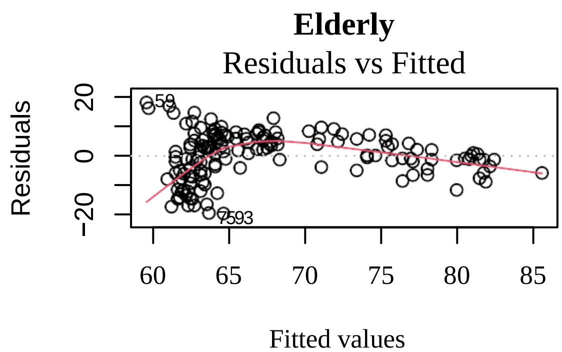

```r
#RURAL
rural_model= lm(life_expect~rural)
plot(rural_model,which=1,main='Rural')
plot(rural_model,which=2,main='Rural')

#HEALTH
health_model= lm(life_expect~health)
plot(health_model,which=1,main='Health')
plot(health_model,which=2,main='Health')
```

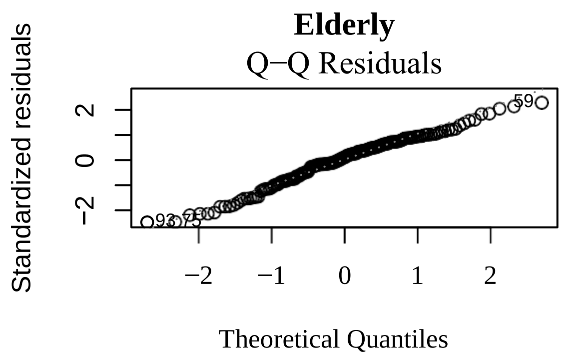



```
#INTERNET
internet_model=         lm(life_expect~internet)
plot(internet_model,which=1,main=              'Internet')
plot(internet_model,which=2,main=              'Internet')

#ELDERLY
elderly_model=        lm(life_expect~elderly)
plot(elderly_model,which=1,main=             'Elderly')
plot(elderly_model,which=2,main=             'Elderly')
```

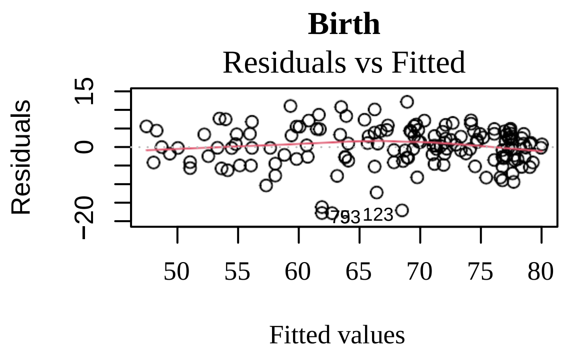


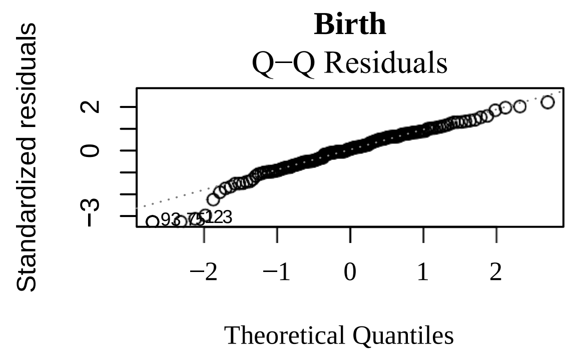


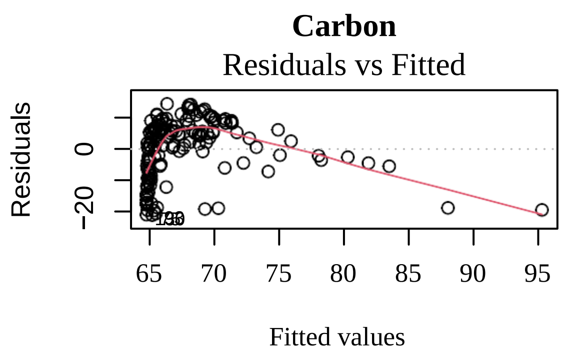


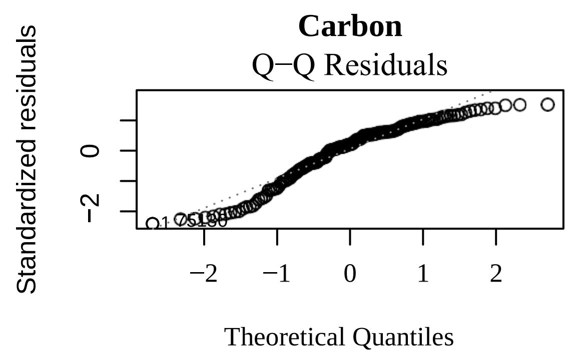



```
#BIRTH
birth_model=     lm(life_expect~birth)
plot(birth_model,which=1,main=     'Birth')
plot(birth_model,which=2,main=     'Birth')

#CARBON
carbon_model=     lm(life_expect~carbon)
plot(carbon_model,which=1,main=     'Carbon')
plot(carbon_model,which=2,main=     'Carbon')
```

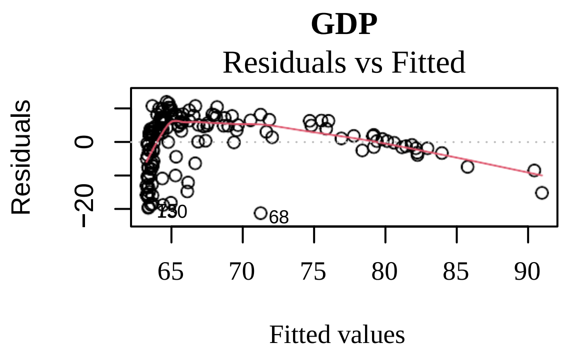


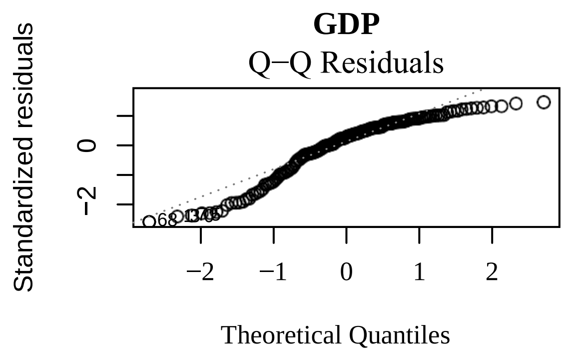


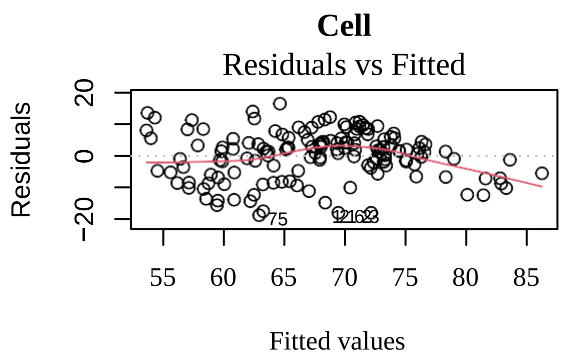


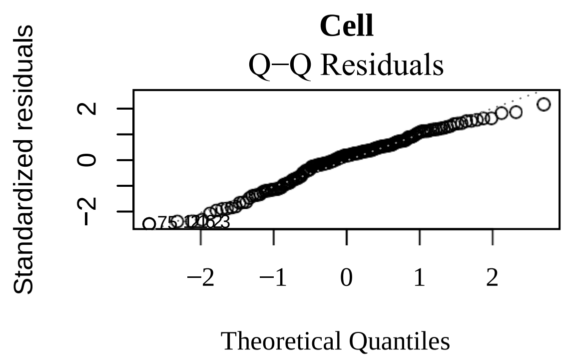



```r
#GDP
GDP_model= lm(life_expect~GDP)
plot(GDP_model,which=1,main= 'GDP')
plot(GDP_model,which=2,main= 'GDP')

#CELL
cell_model= lm(life_expect~cell)
plot(cell_model,which=1,main= 'Cell')
plot(cell_model,which=2,main= 'Cell')
```









Almost all plots show a deviation from normality and/or nonconstant errors, indicating a transformation in Y is needed. The variables: area, pop, internet, elderly, carbon, GDP, and cell all show signs of nonlinearity, indicating a transformation in X is also needed.

Initial Multiple Linear Regression

```r
#First look at which relationships are significant using t-test
overall_model=    lm(life_expect~area+pop+rural+health+internet+elderly+birth+carbon+GDP+cell)
summary(overall_model)
```

```
Call:
lm(formula = life_expect ~ area + pop + rural + health + internet + elderly + birth + carbon
+ GDP + cell)
Residuals:
Min     1Q   Median     3Q     Max
 -16.4565  -2.3102  0.1329  3.1021  11.0534
Coefficients:
```

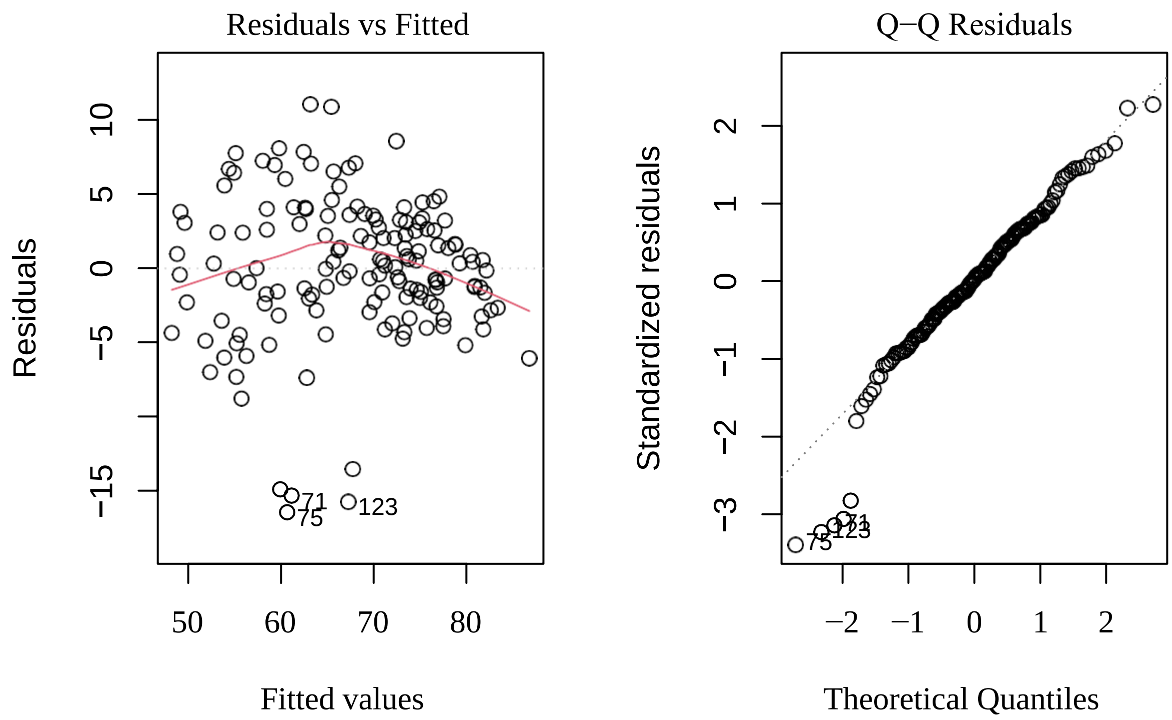
	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	8.108e+01	3.470e+00	23.362	< 2e-16 ***
area	-2.933e-07	2.399e-07	-1.223	0.22352
pop	1.008e-03	4.386e-03	0.230	0.81853
rural	-2.069e-02	2.633e-02	-0.786	0.43334
health	1.202e-01	1.119e-01	1.074	0.28459
internet	9.961e-02	3.560e-02	2.798	0.00588 **

elderly	-4.164e-01	1.632e-01	-2.551	0.01184 *
birth	-6.857e-01	7.749e-02	-8.849	3.96e-15 ***
carbon	-1.256e-01	8.147e-02	-1.541	0.12559
GDP	5.059e-05	4.974e-05	1.017	0.31089
cell	2.826e-02	1.354e-02	2.086	0.03879 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 4.94 on 137 degrees of freedom
Multiple R-squared:0.7881, Adjusted R-squared: 0.7727
F-statistic: 50.96 on 10 and 137 DF, p-value: < 2.2e-16

According to the above p-values, internet, elderly, birth, and cell are significant predictors of life expectancy.

```
#More regression diagnostics to see if violations were corrected
par(mfrow=c(1,2))
plot(overall_model,which=1)
plot(overall_model,which=2)
```



Indications of nonlinearity, nonconstant variance, and non-normality still exist. Transform variables.

Y Transformations

```
#Use Box-Cox Method to evaluate best y-transformation
par(mfrow=c(2,5))

#bc_area = boxcox(area_model, lambda=seq(-10,10,1/10), plotit=F)
```

```
bc_pop=     boxcox(pop_model,lambda= seq(-7,7,1/10),plotit=F) bc_rural=
            boxcox(rural_model,lambda= seq(-7,7,1/10),plotit=F)
bc_internet=    boxcox(internet_model,lambda=seq(-7,7,1/10),plotit=F) bc_health=
               boxcox(health_model,lambda= seq(-7,7,1/10),plotit=F) bc_elderly=
               boxcox(elderly_model,lambda= seq(-7,7,1/10),plotit=F) bc_birth=
            boxcox(birth_model,lambda= seq(-7,7,1/10),plotit=F) bc_carbon=
            boxcox(carbon_model,lambda= seq(-7,7,1/10),plotit=F) bc_GDP=
            boxcox(GDP_model,lambda= seq(-7,7,1/10),plotit=F)
bc_cell=     boxcox(cell_model,lambda= seq(-7,7,1/10),plotit=F)

#bc_area$[which.max(bc_area$y)]

cbind(bc_pop$x[which.max(bc_pop$y)], bc_rural$x[which.max(bc_rural$y)],
        bc_internet$x[which.max(bc_internet$y)], bc_health$x[which.max(bc_health$y)],
        bc_elderly$x[which.max(bc_elderly$y)])
```

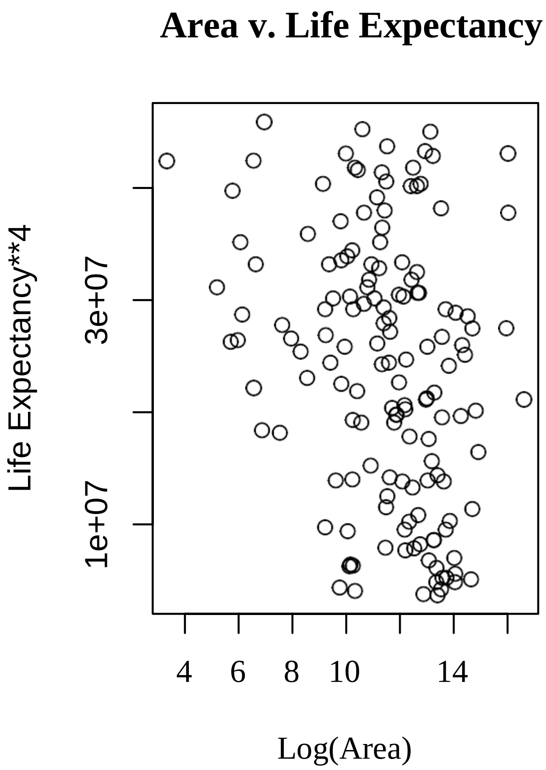
```
[,1] [,2] [,3] [,4] [,5]
[1,]3.2     3.9    5.3    3.5    4
```

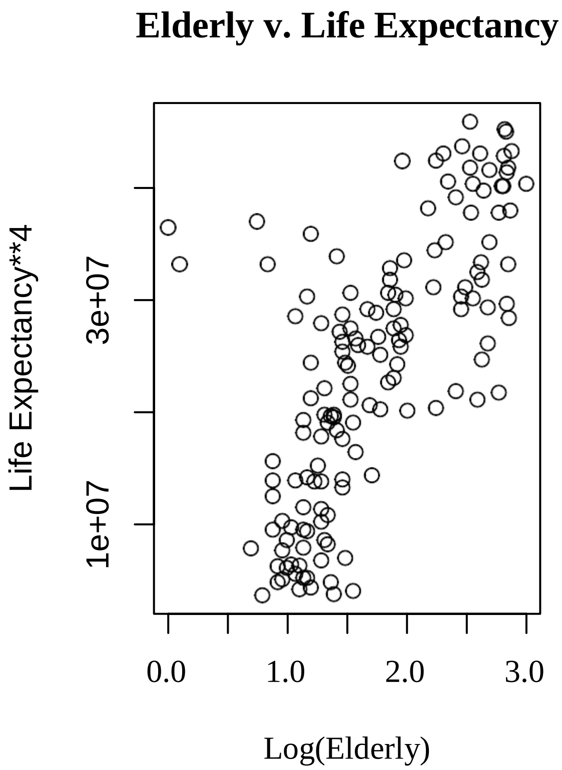
As we can see, there is a common pattern of the range of lambda values that fall within the 95% confidence interval. Therefore, I will choose a lambda=4 since it is a value that occurs often.

X Transformations

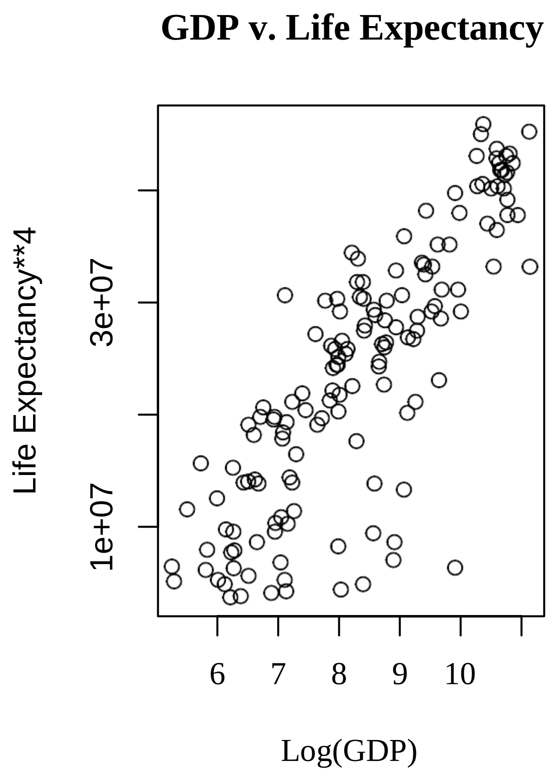
Transformations were found through trial-and-error. Examples of their transformed relationships with our transformed life expectancy are shown below.

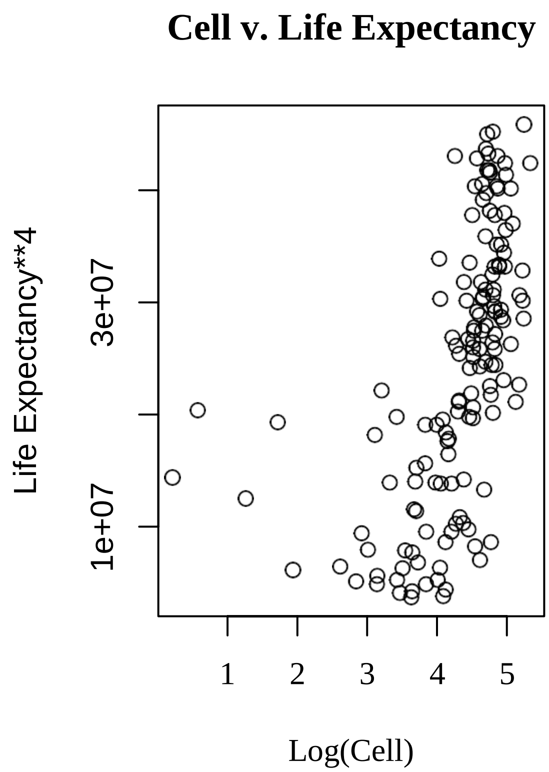
```
par(mfrow=c(1,2))

plot(log(area), life_expect**4,xlab= 'Log(Area)',ylab= 'Life Expectancy**4', main='Area v. Life Expectancy')
plot(log(elderly), life_expect**4,xlab= 'Log(Elderly)',
      ylab='Life Expectancy**4',main= 'Elderly v. Life Expectancy')
```






```
plot(log(GDP), life_expect**4,xlab= 'Log(GDP)',ylab= 'Life Expectancy**4', main='GDP v. Life
        Expectancy')
plot(log(cell), life_expect**4,xlab= 'Log(Cell)',ylab= 'Life Expectancy**4', main='Cell v. Life Expectancy')
```





Transformed Multiple Linear Regression

```
new_LE=life_expect**4
new_area=log(area)
new_pop=log(pop)
new_internet=internet**(1/2)
new_elderly=log(elderly)
new_GDP=log(GDP)
new_cell=log(cell)
```

```
#Throw transformed variables into new MLR to see which predictors are significant now
transformed_model=lm(new_LE~new_area+new_pop+rural+new_internet+health+
                     new_elderly+birth+new_GDP+new_cell)
summary(transformed_model)
```

```
Call:
lm(formula = new_LE ~ new_area + new_pop + rural + new_internet +health +
new_elderly + birth + new_GDP + new_cell)
Residuals:
Min      1Q   Median     3Q      Max
 -15396594 -2786981 -203958  3407028  13838525
Coefficients:

                Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept)     23010024    7835143    2.937    0.00389  **
new_area         -879946     287339   -3.062    0.00264  **
new_pop           820087     369060    2.222    0.02790  *
rural             -35053      28364   -1.236    0.21862
new_internet     1474187     356962    4.130   6.26e-05  ***
health            160069     114099    1.403    0.16289
new_elderly       -42689    1135620   -0.038    0.97007
birth            -409824      87608   -4.678   6.83e-06  ***
new_GDP          1417013     648873    2.184    0.03067  *
new_cell           93080     659389    0.141    0.88795
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 5018000 on 138 degrees of freedom
Multiple R-squared:  0.8325, Adjusted R-squared:  0.8215
F-statistic: 76.19 on 9 and 138 DF, p-value: < 2.2e-16
```

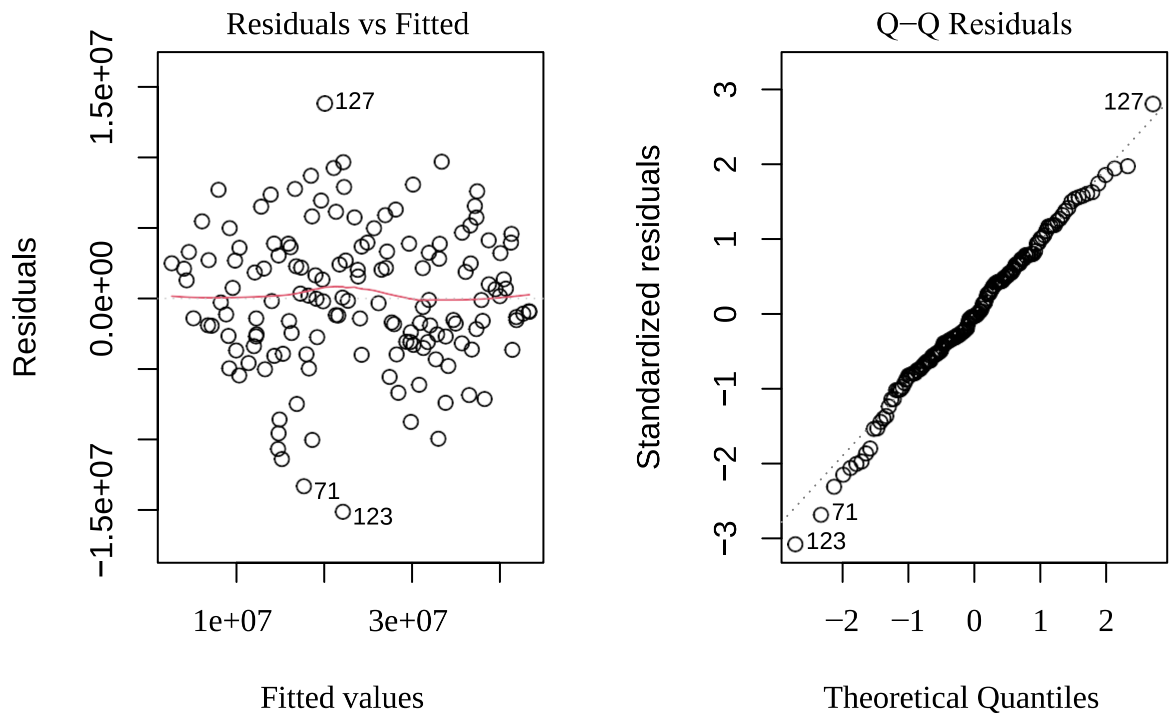
Now, according to our p-values, area, population, internet, birth, and GDP are significant predictors of life expectancy.

Final Model

```
#Attempt model with only important variables
final_model=  lm(life_expect**4~new_area+new_pop+new_internet+birth+new_GDP)
summary(final_model)
```

```
Call:
lm(formula = life_expect^4 ~ new_area + new_pop + new_internet + ##     birth +
new_GDP)
Residuals:
     Min       1Q   Median      3Q      Max
-15089880 -3064764 -134918  3297799  13817901
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    18420514    5403623   3.409 0.000849 ***
new_area        -826711     284426  -2.907 0.004241  **
new_pop          751206     360547   2.084 0.038997  *
new_internet    1675575     322733   5.192  7.07e-07  ***
birth           -390819      67880  -5.757  5.07e-08  ***
new_GDP         1804916     560596   3.220 0.001591  **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 5008000 on 142 degrees of freedom
Multiple R-squared:  0.8283, Adjusted R-squared:0.8222
F-statistic:137 on 5 and 142 DF, p-value: < 2.2e-16
```

```
par(mfrow=c(1,2))
plot(final_model,which=1)
plot(final_model,which=2)
```



Residual and QQplot show no strong violations of our assumptions. All of our predictors are significant.

Model Selection

```
#Use forward/backward selection to understand most important predictors
forward_LEmodel=     regsubsets(new_LE~new_area+new_pop+new_internet+birth+
                                    new_GDP,data=count_subset,method="forward")
summary(forward_LEmodel)
```

```
Subset selection object
Call: regsubsets.formula(new_LE ~ new_area + new_pop + new_internet + ## birth +
new_GDP, data = count_subset, method = "forward")
5 Variables          (and intercept)
                     Forced   in   Forced   out
new_area      FALSE     FALSE
new_pop                FALSE            FALSE
new_internet  FALSE    FALSE
birth                   FALSE      FALSE
new_GDP        FALSE     FALSE
1 subsets of each size up to 5 ## Selection
Algorithm: forward
        new_area       new_pop   new_internet       birth    new_GDP
1       ( 1 ) " " " "       "*"       " "               " "
2       ( 1 ) " "           " "       "*"               "*"      " "
3       ( 1 ) " "           " "       "*"               "*"      "*"
4       ( 1 ) "*"           " "       "*"               "*"      "*"
5       ( 1 ) "*"           "*"       "*"               "*"      "*"
```

Backward model selection shows the same results.

According to forward selection, the most significant variables in order from most significant to least significant are: internet, birth, GDP, area, and population.

```
#Use LEAPS to do model selection to pick a parsimonious model
library(leaps)
design_final=    cbind(new_area, new_pop, new_internet, birth, new_GDP)
```

```
#try it with adjusted R2
adjr2_model=    leaps(x=design_final,y=new_LE,method=      c('adjr2'),
                     names=c('Area','Population','Internet','Birth','GDP'))
adjr2_model$adjr2
```

```
 [1] 0.726358152 0.702230338 0.684551562 0.070835858 0.004828808 0.804816576
 [7] 0.777720664 0.765348990 0.738928449 0.727276815 0.705900683 0.700491715
[13] 0.693500283 0.682776915 0.093910913 0.814170905 0.809316803 0.804212790
[19] 0.780584413 0.776364362 0.773879336 0.763967551 0.741346056 0.716607271
[25] 0.708139311 0.818089341 0.812983820 0.810599851 0.789978617 0.782279844
[31] 0.822242449
```

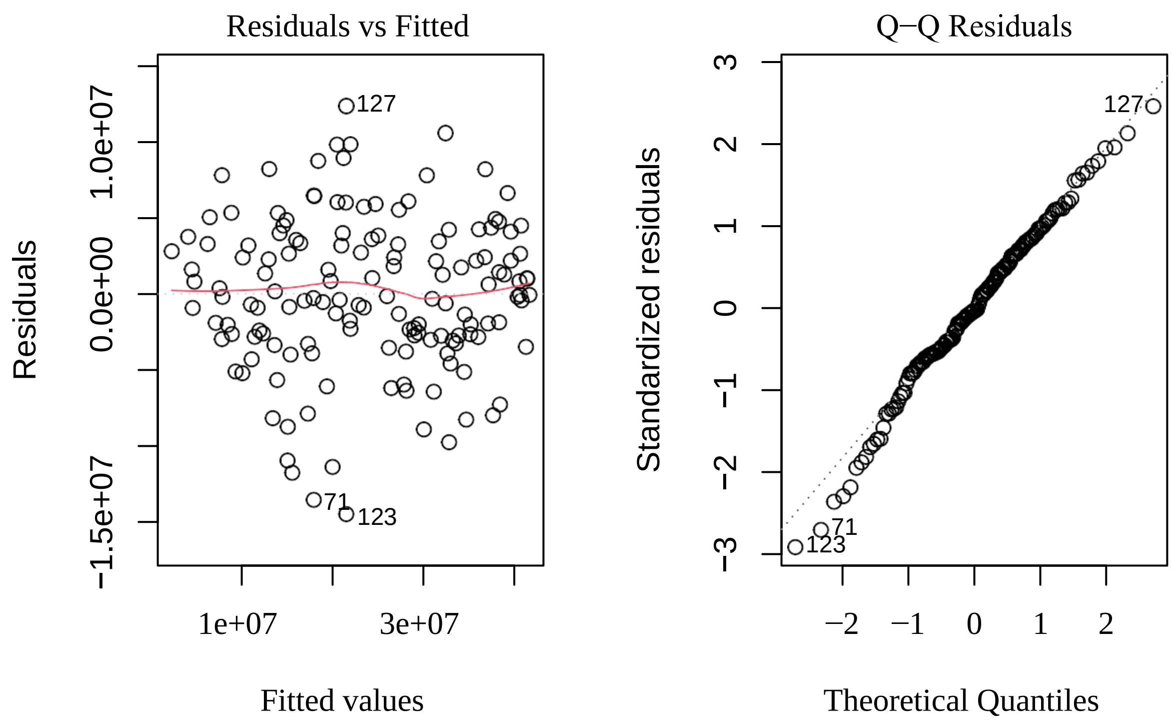
The highest parisimonious adjusted R2 is with population excluded.

```
#Try with Mallow's
mallow_LEmodel=   leaps(x=design_final,y=new_LE,method=      c('Cp'),
                       names=c('Area','Population','Internet','Birth','GDP'))
mallow_LEmodel$Cp
```

```
 [1]  80.753939 100.571161  115.091508  619.162881  673.377340   17.214595
 [7]  39.317213  49.409008   70.960713   80.465159   97.902051  102.314242
[13] 108.017278 116.764520  597.113005   10.538695   14.470965   18.605686
[19] 37.746849   41.165479   43.178579   51.208039   69.533535   89.574230
[25] 96.434059    8.341037   12.448257   14.366080   30.955173   37.148578
[31] 6.000000
```

Best parsimonious value of Mallow's Cp is again with population excluded.

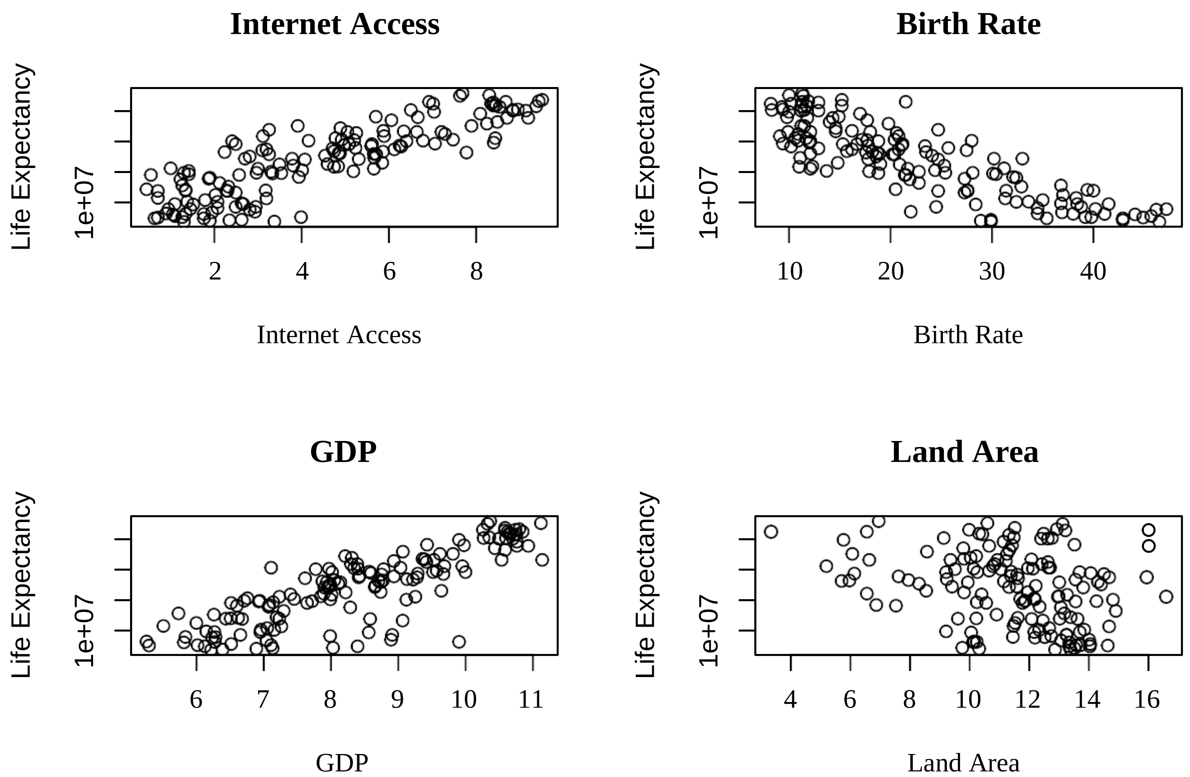
```
#final model results
final_model=    lm(new_LE~new_internet+birth+new_GDP+new_area)
par(mfrow=c(1,2))
plot(final_model,which=1) plot(final_model,which=2)
```




```
standard_model=       lm(scale(new_LE)~scale(new_internet)+scale(birth)+
                          scale(new_GDP)+scale(new_area))
confint(standard_model)
```

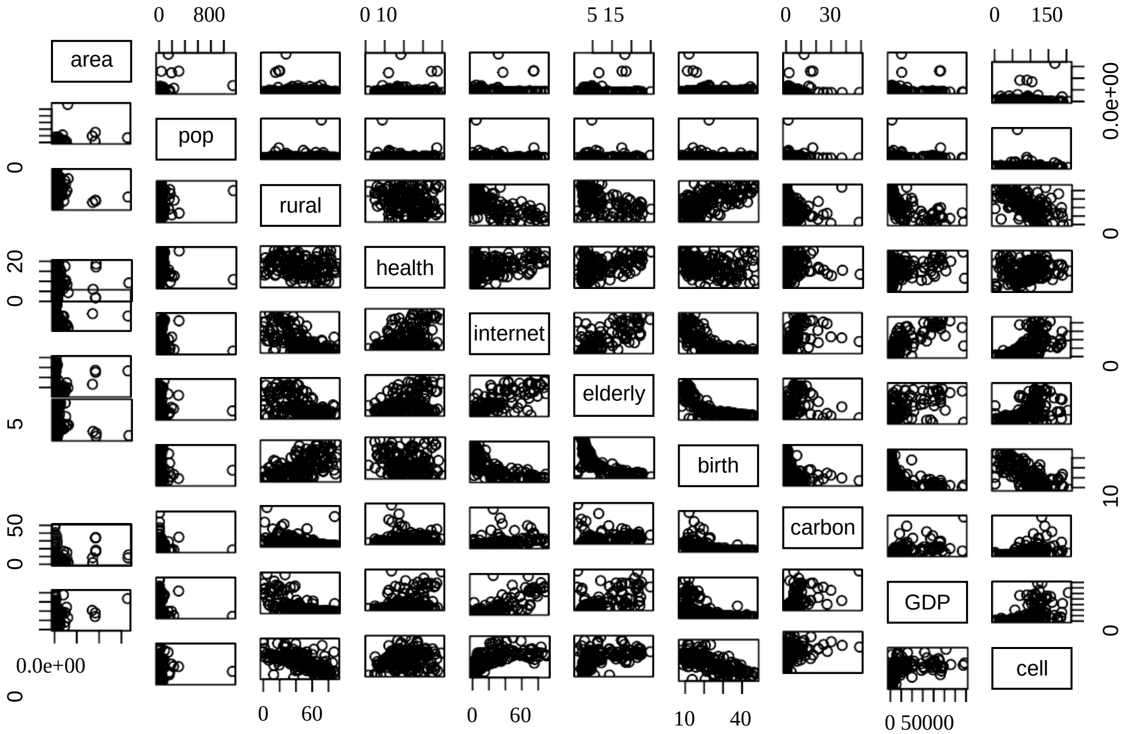
	2.5 %	97.5 %
(Intercept)	-0.06930064	0.069300643
scale(new_internet)	0.24977023	0.526902541
scale(birth)	-0.48325676	-0.243737786
scale(new_GDP)	0.05989644	0.341165992
scale(new_area)	-0.14532020	-0.001764897

```
par(mfrow=c(2,2))
plot(new_internet, new_LE,xlab= 'Internet Access',ylab= 'Life Expectancy', main='Internet
     Access')
plot(birth, new_LE,xlab= 'Birth Rate',ylab= 'Life Expectancy', main='Birth Rate')
plot(new_GDP, new_LE,xlab= 'GDP',ylab= 'Life Expectancy', main='GDP')
plot(new_area, new_LE,xlab= 'Land Area',ylab= 'Life Expectancy', main='Land Area')
```



Multicollinearity

```
#Check for multicollinearity for all variables
pairs(cbind(area,pop,rural,health,internet,elderly,birth,carbon,GDP,cell))
```



#Displaying correlation matrix of all the predictors

```
design = cbind(life, area, pop, elderly,cell,rural,health,birth,internet,carbon,gdp)
cor(design)
```

(Below result is the edited version of the matrix to accommodate all the values)

	life	area	popu	eldpopu	cell	rural	health	birth	internet	carbon	gdp
life	1	0.00718	-0.0249	**0.641**	**0.685**	**-0.621**	0.265	**-0.850**	**0.7414**	0.431	**0.617**
area		1	0.342	0.0885	0.047	-0.131	0.00749	-0.067	**0.741**	0.431	**0.616**
popu			1	-0.013	-0.0639	0.0636	-0.1445	-0.0115	0.077	0.128	0.077
elderly				1	0.495	-0.495	0.393	**-0.75**	-0.039	-0.034	-0.036
cell					1	-0.60	0.197	-0.684	**0.745**	0.205	0.56
rural						1	-0.191	0.599	0.597	0.443	0.49
health							1	-0.1996	-0.665	-0.43	-0.649
birth								1	0.395	0.027	0.343
internet									1	0.4855	**0.804**
carbon										1	0.592
gdp											1

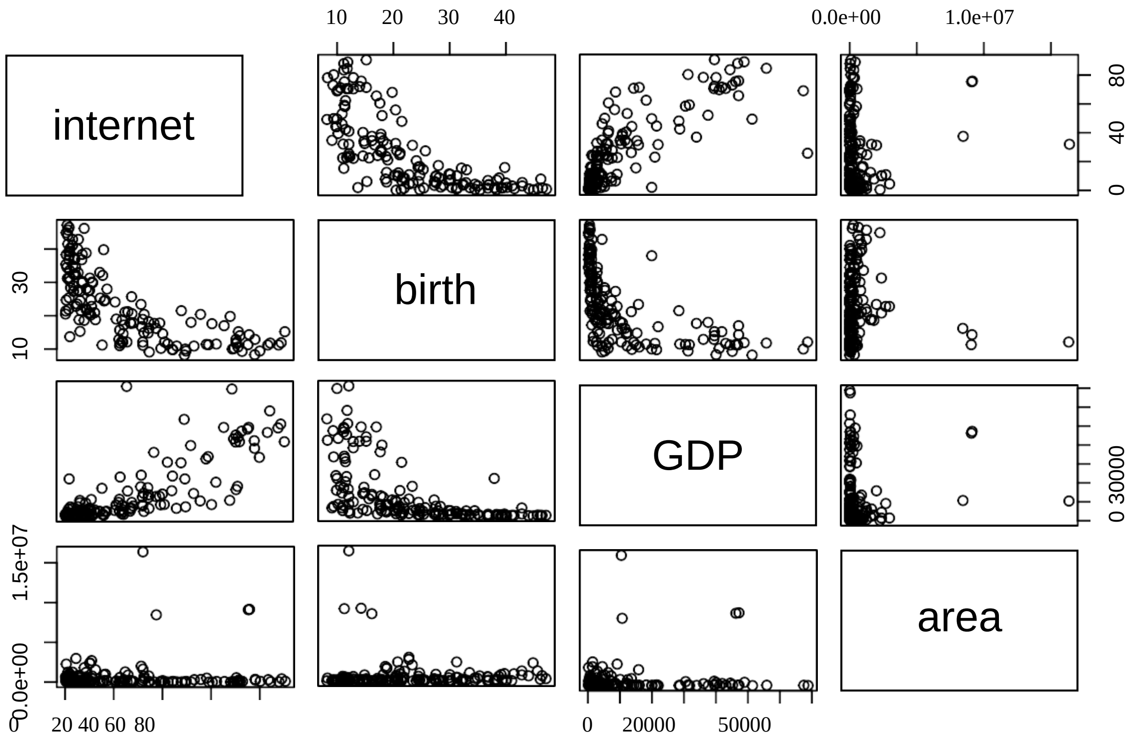
#Displaying Variance Inflation Factor (VIF) of all the predictors

```
library(faraway)
vifmodel <- lm(life ~ area + health + carbon + elderly + popu + rural + cell + birth + gdp + internet)
vif(vifmodel)
```

```
    area   health   carbon  elderly     popu    rural     cell    birth      gdp internet
1.203130 1.365423 2.063836 3.788557 1.197367 2.302427 2.208491 4.037504 3.736785 5.051648
```

Internet, GDP, Birth rate are moderately collinear but Area is less collinear with other predictors.

```
#Check for multicollinearity for selected predcitors
pairs(cbind(internet,birth,GDP,area))
```



Multicollinearity exists between internet and birth, birth and GDP, and internet and GDP.

```
#Displaying correlation matrix of the selected predictors
Design1 = cbind(internet,birth,gdp,area)
cor(design1)
```

	internet	birth	gdp	area
internet	1.00000000	-0.71589970	0.80441383	0.07777323
birth	-0.71589970	1.00000000	-0.57035251	-0.06777425
gdp	0.80441383	-0.57035251	1.00000000	0.07770053
area	0.07777323	-0.06777425	0.07770053	1.00000000

```
#Displaying Variance Inflation Factor (VIF) of the model predictors
vifmodel1 <- lm(life ~ area + birth + gdp + internet)
vif(vifmodel1)

    area    birth      gdp internet
1.007060 2.052343 2.835907 3.922992
```

Before transformations, Internet has the highest VIF value of 3.92, followed by GDP, birth rate and area. Area is the least correlated and Internet is moderately correlated with other model predictors.

```
#Displaying Variance Inflation Factor (VIF) of the transformed model predictors
vifmodel1 <- lm(life^4 ~ log(area) + birth + log(gdp) + sqrt(internet))
vif(vifmodel1)

log(area)      birth   log(gdp) sqrt(internet)
1.065515  2.966203  4.090407  3.970959
```

After transformation, GDP has the highest VIF value of 4.09, which is not so different from that of Internet (3.97), and followed by birth rate and area. Hence, GDP and Internet are moderately correlated whereas birth rate is less correlated, and area is the least correlated.